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Business and government applications of text mining & Natural Language Processing (NLP) for societal benefit: Introduction to the special issue on "text mining & NLP"

Natural Language Processing (NLP) is a branch of artificial intelligence (AI) that helps computers to understand, process, and analyze large amounts of natural human language data [4]. The concepts of AI, machine learning (ML), and human-to-machine interactions have been prevalent for several decades. A growing number of businesses have used advanced analytics and ML methods for resolving their business concerns, and NLP is emerging as one of the most popular and indispensable technologies for organizations today. Given the complexities of NLP and other AI models, explainable AI has also become more important in both research and practice domains [6].

NLP includes natural human communication variables including text, audio, and video, and its modeling, interpretations and applications. Text mining, a component of NLP, deals with the analysis of textual datasets to discover novel and useful patterns and trends, while speech models can capture, interpret, translate, and transcribe speech, and perform other emerging functions. Different components of NLP can be used synchronously for achieving different goals, such as using audio and video together to classify and categorize videos on social media sites. Common business applications that use text mining include social media monitoring, where businesses rely on these technologies to understand the sentiments (i.e., mood and the emotions) of customers by analyzing user generated data [4]. Smart use of NLP empowers businesses to gain a competitive edge over their rivals in modern market spaces. According to a recent estimate, around 80% of business data is non-actionable due to being unstructured in nature [1]. As the unstructured data volume overloads several sectors in business and government (e.g., finance, e-commerce, healthcare, hospitality sectors in business, and business classification, request for proposals (RFP), trust and confidence, public comments in government areas), NLP and other AI technologies are becoming increasingly crucial for business growth and survival. Several emerging NLP business applications have the ability to revamp and revitalize struggling businesses, by demystifying obscure statistical business reports into precise actionable data, leading to streamlined action plans and higher revenues in the long run.

One of the biggest challenges faced by businesses is to understand human behaviors. Machine learning and text mining have enabled e-commerce platforms to extract attributes and hence improve product searches to target the right customers. Similarly, NLP has played a vital role in improving the performance of healthcare systems, identify harmful content and model cybersecurity defences.

The six timely and interesting papers of the special issue provide insights into these and other domains, and showcase their modeling using NLP and text mining tools to solve novel research problems in businesses, government and society. While each paper contributes to the literature with insights for researchers and practical applications for managers, they also provide future research directions in NLP and text analytics modeling and application domains. We provide a synopsis of each accepted paper of the special issue below, and invite researchers to study and contribute to this emerging and promising area of research in NLP and text mining for societal benefit.

In "*Walking on air or hopping mad? Understanding the impact of emotions, sentiments and reactions on ratings in online customer reviews of mobile apps*", authors Dhar and Bose study an important problem in business – of understanding the emotions of customers in online reviews. They conduct an analysis of online customer reviews (OCR) from mobile payment applications and develop a perception score using text & emojis. They use this NLP model to capture the degree of OCR more effectively. Authors use sentiment analysis to determine whether ratings of mobile payment applications capture the customer perception reflected in the text data of the OCR. They show that results from this paper will be useful for mobile app developers that use online customer reviews to better market their applications and influence future downloads. Theoretical contributions include analyzing the association between opinions expressed in online consumer reviews and start ratings for the mobile app industry, using a three-dimensional data mining approach, to show that it is important to capture sarcasm embedded in customer reviews. This research can be extended by studying apps on other mobile platforms, payment apps in different geographical locations, and apps providing other digital services. Further research is needed to test the usability and acceptability of the perception score in lieu of, or in addition to, start ratings.

"*Understanding Consumer Engagement in Social Media: The Role of Product Lifecycle*" studies customer engagement in social media based on user participation and content co-creation. Authors Eslami et al. use social media data from multiple platforms, and develop an NLP model to investigate the effects of popularity and argument frame to study customer engagement. They find that a product promoter's popularity in social media is a significant determinant of increasing customer engagement, as is an objectively written post, while the number of comments are not a significant contributor. Authors also find a moderating effect of product lifecycle on such customer engagement. The findings of this study aim to help firms make better decision to enhance their customer engagement on different social media channels. Future research includes studying the generalizability of these findings across other social media platforms, and also for different products and service types, cultures and languages.

Kumar et al. utilize NLP and computing intelligence to model important psychological factors in healthcare during the COVID-19

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pandemic. In "*Mining Voices from Self-expressed Messages on Social-Media: Diagnostics of Mental Distress during COVID-19*", they try to identify factors that could affect mental health, and could be precursors to depression and suicidal tendencies in people. Authors observed that lockdown anxiety during COVID-19 emerged as a major factor leading to poor mental health, and found that psychological disorders have remained pre-eminent. Authors find that financial burden was a major cause before COVID-19 which led to high odds of suicidal thoughts in people, and led to depressive thoughts for those who lost jobs in the pandemic. The research provides implications and a roadmap for health policy makers, psychologists, government and practitioners, to employ various preventive and corrective measures to fight mental stress in individuals during a pandemic situation. This research can be further augmented in future by studying patterns as lockdowns ease across the world, and new challenges emerge such as work-life balance in the post COVID-19 pandemic workspace, as well as the performance of new graduate post pandemic hires in the workplace.

"*Impact of information timeliness and richness on public engagement on social media during COVID-19 pandemic: An empirical investigation based on NLP and machine learning*", by authors Li et al., examines the impact of information timeliness and richness on public engagement. The paper uses text data from China's largest social media platform during times of the COVID-19 pandemic, using a similarity calculation method based on NLP and text mining. The findings show that information retrospectiveness is negatively correlated with public engagement breadth but positively with depth. In addition, the findings indicate that both information immediateness and prospectiveness improved the breadth and depth of public engagement. Moreover, information richness has a positive moderating effect on the relationships between information retrospectiveness, prospectiveness, and public engagement breadth but its impact on immediateness is not significant. While the paper has a good contribution in the context of social media, future studies could examine the impact of other dimensions of information quality on individuals' engagement. In addition, the focus of this paper is limited to the health context. Future studies could focus on other contexts.

"*A Multistate Modeling Approach for Organizational Cybersecurity Exploration and Exploitation*" studies organizational cyber security responses using NLP and Markov models. Authors Zadeh and Jeyaraj analyze textual data of 10-K reports of S&P 100 companies using text analytics, and employ multistate modeling to examine the dynamic stages of exploration and exploitation efforts in their cybersecurity responses. Authors use a combination of NLP and two-state Markov model to quantify the deeper levels of exploration and exploitation and consider different states e.g., investigating, balancing, surviving and reinforcing in the cybersecurity responses of organizations to analyze the dynamics of exploration-exploitation behaviors. Authors provide research and practical implications of their research, and suggest that organizations need to maintain their long-term focus on both exploration and exploitation to achieve ambidexterity in organizational cybersecurity. Further, organizations need to engage in incremental and gradual transitions between varying levels of exploration and exploitation for survival and competitive advantage. Future research in cybersecurity modeling can consider other primary and benchmark data sources, and utilize alternative modeling approaches such as game theory.

In "*Mining longitudinal user sessions with deep learning to extend the boundary of consumer priming*", Cheng and Chen use the NLP and deep learning models and propose a two-stage product evaluation process to extend the boundary of priming in an online competitive environment. Authors use the shortlisting stages into product comparison and validate the model of priming process on a B&B booking platform. Authors analyze the review text data and long-term tracking sessions and show that there are opportunities available for the marketing managers to use the NLP method to begin an effective priming process for the businesses.

In conclusion, we are happy to highlight and share these novel and high-quality research articles in data science and business analytics, using NLP, text mining, big data analytics and related technologies including machine learning and deep learning, which contribute significantly to the research literature, and are also directly applicable to businesses, government and society [2–5].

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